INFORMATION-THEORETIC PROPERTIES OF PRIME GAP SEQUENCES

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Abstract. We study the prime gap sequence $g_n = p_{n+1} - p_n$ through the lens of information theory. Our main result is a comparison of the empirical mutual information $\mathrm{MI}(g_n; g_{n+1}) \approx 0.33$ bits against the Cramér random model, under which consecutive gaps are independent (so MI = 0). The full 0.33 bits of lag-1 dependence is therefore attributable to the multiplicative structure of the integers—the sieve. We verify this via permutation tests ($z > 1000$ at lag 1; lag 2 also significant; lags ≥ 3 indistinguishable from noise) and show that the non-Gaussian discrete character of prime gaps makes the mutual information approximately 300 times larger than the Gaussian prediction $\rho^2/(2\ln 2)$ from the linear autocorrelation $\rho_1 \approx -0.04$. Supporting experiments include: block entropy measurements with an explicit finite-sample ceiling analysis showing H_k saturates at $\log_2 N$ for large k; a Lempel–Ziv entropy rate estimator that avoids this curse of dimensionality, yielding $h_{\mathrm{LZ}}^{\mathrm{gaps}}/h_{\mathrm{LZ}}^{\mathrm{shuffled}} \approx 0.92$–$0.94$; goodness-of-fit tests confirming monotonic convergence toward the Gallagher–Poisson prediction with variance-to-mean ratio climbing from 0.72 to 0.77; and conditional entropy measurements revealing a $\sim$0.06-bit asymmetry between primes $\equiv 1$ and $\equiv 5 \pmod 6$. We also verify that the empirical single-gap entropy satisfies $H_1 \approx \log_2(\log x) + C$ with $C \approx 0.44$, consistent with a discretized exponential model weighted by the Hardy–Littlewood singular series. All code and data are publicly available.

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1. Introduction

The prime gap sequence $g_n = p_{n+1} - p_n$ is one of the most studied objects in analytic number theory. Classical results characterize its average behavior (the prime number theorem gives $\langle g_n \rangle \sim \log p_n$), its extreme values (the Cramér conjecture [1], the Maynard–Tao theorem on bounded gaps [3]), and its distributional properties (the Gallagher conjecture [2] predicting local Poisson count statistics conditional on the Hardy–Littlewood k-tuple conjecture). Lemke Oliver and Soundararajan [4] showed that consecutive primes exhibit unexpected biases in their joint distribution of residues modulo q, traceable to the Hardy–Littlewood singular series.

Despite this extensive literature, the *information-theoretic* properties of the gap sequence have received limited systematic attention. Shannon entropy, mutual information, and algorithmic complexity provide a complementary lens that captures aspects of the sequence's structure—including nonlinear dependencies and compressibility—that linear statistical measures like autocorrelation cannot detect. Cohen, Iyer, and Manack [5] recently studied the exponential fit of normalized prime gaps, providing a distributional perspective that our work complements with information-theoretic tools.

The central contribution of this paper is a *theoretical prediction*: under the Cramér random model, consecutive prime gaps are independent (Proposition 3.1), so $\mathrm{MI}(g_n; g_{n+1}) = 0$. The empirical value of ≈ 0.33 bits, verified via permutation tests, is therefore entirely a signature of the multiplicative sieve. We quantify this sieve contribution and compare it to predictions from the Hardy–Littlewood singular series.

Remark 1.1 (Stationarity)**.** The prime gap sequence is not stationary: the gap distribution changes with scale as $\langle g_n \rangle \sim \log p_n$. Throughout this paper, we treat each fixed-scale window of $N = 2 \times 10^5$ consecutive primes as approximately stationary. All entropy-rate and mutual-information measurements are made within such windows; cross-scale comparisons reflect the evolution of the local gap distribution, not a single stationary process.

2. Definitions and Setup

Let $p_1 = 2 < p_2 = 3 < p_3 = 5 < \cdots$ be the primes. Fix a starting scale x and let $p_1(x) < p_2(x) < \cdots < p_N(x)$ be N consecutive primes starting near x. Define the gap sequence $g_i = p_{i+1}(x) - p_i(x)$ for $i = 1, \ldots, N-1$.

Definition 2.1 (Block entropy)**.** For $k \geq 1$, define the k-block entropy

$$H_k = -\sum_{\mathbf{w} \in \mathcal{A}^k} \hat{p}(\mathbf{w}) \log_2 \hat{p}(\mathbf{w}),$$

where $\hat{p}(\mathbf{w})$ is the empirical frequency of the k-tuple $\mathbf{w} = (w_1, \ldots, w_k)$ among the overlapping windows $(g_i, g_{i+1}, \ldots, g_{i+k-1})$, and $\mathcal{A} = \{2, 4, 6, 8, \ldots\}$ is the alphabet of even positive integers.

Definition 2.2 (Entropy rate)**.** The k-th entropy rate estimate is $h_k = H_k - H_{k-1}$ for $k \geq 2$, with $h_1 = H_1$. For a stationary ergodic process, $h_k \to h$ as $k \to \infty$, where h is the entropy rate of the process.

For an i.i.d. process, $H_k = k \cdot H_1$ and $h_k = H_1$ for all k. The *redundancy* $R_k = k \cdot H_1 - H_k \geq 0$ measures the total dependence captured up to block size k. Note that R_k equals the *total correlation* (multi-information) of the k-tuple $(g_n, \ldots, g_{n+k-1})$.

Definition 2.3 (Mutual information)**.** For random variables X, Y with discrete joint distribution,

$$\mathrm{MI}(X; Y) = H(X) + H(Y) - H(X, Y).$$

We apply the Miller–Madow bias correction [7]: subtract $(|\mathcal{A}_{XY}| - |\mathcal{A}_X| - |\mathcal{A}_Y| + 1)/(2n \ln 2)$ from the raw plug-in estimate, where $|\mathcal{A}_{XY}|$ is the number of observed joint categories.

Definition 2.4 (Lempel–Ziv complexity and entropy rate)**.** The LZ76 complexity $C(s)$ of a finite sequence $s = s_1 s_2 \cdots s_n$ is the number of distinct phrases in the sequential parsing of s into shortest substrings not previously encountered [6]. For a stationary ergodic process over an alphabet of size $|\mathcal{A}|$,

$$\frac{C(s) \cdot \log_{|\mathcal{A}|} n}{n} \to h \quad \text{almost surely as } n \to \infty,$$

where h is the entropy rate [6]. We use this as a *consistent* entropy rate estimator that avoids the curse of dimensionality inherent in plug-in block entropy for large k.

3. The Cramér Model and Mutual Information

We begin with the paper's main theoretical result: a comparison of the empirical lag-1 mutual information to the prediction of the Cramér random model.

Proposition 3.1 (Independence under the Cramér model)**.** *In the Cramér random model near x — where each integer $m > p$ is independently "prime" with probability $1/\log x$ — consecutive gaps g_n and g_{n+1} are independent. Consequently,* $\mathrm{MI}(g_n; g_{n+1}) = 0$ *under this model.*

Proof. In the Cramér model with constant density $1/\log x$, the gap $g_n = p_{n+1} - p_n$ is determined by the primality of integers $p_n + 1, p_n + 2, \ldots$, which are independent. Given p_{n+1}, the gap g_{n+1} depends only on the primality of integers $p_{n+1} + 1, p_{n+1} + 2, \ldots$, a disjoint set of integers. Since primality assignments are independent in the Cramér model, g_{n+1} is conditionally independent of g_n given p_{n+1}, and since the density is approximately constant near x, this extends to marginal independence. More precisely, gaps follow a geometric distribution $\Pr(g = k) \approx (1 - 1/\log x)^{k-1}/\log x$ for $k \geq 1$, and by the memoryless property of the geometric distribution, the waiting time beyond p_{n+1} is independent of the waiting time to reach p_{n+1}. □

Remark 3.2. In the Cramér model with varying density ($1/\log m$ depending on m), there is a residual coupling of order $O(1/(\log x)^2)$ because a large g_n shifts p_{n+1} to a region where $1/\log m$ is slightly different. This coupling is negligible at all scales we consider.

Observation 3.3 (The sieve contribution to MI)**.** Since $\mathrm{MI}_{\text{Cramér}} = 0$ and $\mathrm{MI}_{\text{empirical}} \approx 0.33$ bits at lag 1, the entire observed mutual information is attributable to the multiplicative structure of the integers—the sieve. The sieve creates correlations through the Hardy–Littlewood singular series: the probability of a gap of size h depends on the prime factorization of h via $\mathfrak{S}(\{0,h\}) = 2C_2 \prod_{p|h,\, p>2}(p-1)/(p-2)$. Since consecutive gaps $g_n = h_1$ and $g_{n+1} = h_2$ involve three primes $(p_n, p_n + h_1, p_n + h_1 + h_2)$, their joint probability depends on the *triple* singular series $\mathfrak{S}(\{0, h_1, h_1 + h_2\})$, which generally does not factor as $\mathfrak{S}(\{0,h_1\}) \cdot \mathfrak{S}(\{0,h_2\})$. This non-factorization is the source of the mutual information.

4. Block Entropy and the Finite-Sample Ceiling

Table 1 summarizes the block entropy measurements at the largest scale (10^8).

Table 1. Block entropy at scale 10^8 ($N = 200{,}000$ primes).

k	H_k (bits)	h_k (bits)	H_k^{iid} (bits)	n_{distinct}
1	4.481	4.481	4.481	80
2	8.634	4.153	8.963	1,879
3	12.499	3.865	13.444	18,671
4	15.607	3.109	17.926	81,651
5	17.186	1.579	22.407	163,726
6	17.557	0.370	26.889	194,835
7	17.604	0.047	31.370	199,431
8	17.609	0.005	35.852	199,929
9	17.610	0.001	40.333	199,986
10	17.610	0.000	44.815	199,990

Remark 4.1 (Finite-sample ceiling)**.** The saturation of H_k near $\log_2 N \approx 17.6$ bits is primarily a *finite-sample artifact*, not an intrinsic property of the gap process. At $k = 10$, nearly every overlapping window is unique ($n_{\text{distinct}} \approx N$), so the plug-in estimator returns $H_{10} \approx \log_2 N$ regardless of the true entropy rate. The entropy rate h_k appearing to converge to zero for large k reflects this ceiling, not a genuine vanishing of conditional uncertainty. The true entropy rate of the gap sequence (within a fixed-scale window) is almost certainly positive.

The block entropy estimates are *reliable* for small k where $n_{\text{distinct}} \ll N$. In particular:

- At $k = 1$: $n_{\text{distinct}} = 80 \ll 199{,}999$. The estimate $H_1 = 4.481$ bits is well-determined.

- At $k = 2$: $n_{\text{distinct}} = 1{,}879 \ll 199{,}998$. The estimate $H_2 = 8.634$ bits and the drop $h_2 = 4.153 < H_1$ are meaningful, revealing genuine lag-1 dependence.
- At $k \geq 5$: n_{distinct} approaches N, and the estimates begin to be dominated by the sample-size ceiling.

The redundancy $R_2 = 2H_1 - H_2 = 0.33$ bits at $k = 2$ is a robust measurement that matches the lag-1 mutual information (as it must: $R_2 = \text{MI}(g_n; g_{n+1})$).

4.1. **The LZ entropy rate estimator.** To obtain a consistent entropy rate estimate free of the curse of dimensionality, we use the LZ76 complexity. The ratio $h_{\text{LZ}}^{\text{gaps}}/h_{\text{LZ}}^{\text{shuffled}}$ measures the fraction of marginal entropy that survives as true process entropy:

TABLE 2. LZ-based entropy rate estimates. $h_{\text{LZ}} = C \cdot \log_{|\mathcal{A}|}(n)/n$.

Scale	$h_{\text{LZ}}^{\text{gaps}}$	$h_{\text{LZ}}^{\text{shuffled}}$	Ratio
10^4	0.6538	0.7079	0.9235
10^5	0.6580	0.7105	0.9260
10^6	0.6644	0.7152	0.9290
10^7	0.6685	0.7143	0.9359
10^8	0.6825	0.7237	0.9431

The ratio ≈ 0.92–0.94 is the key quantity: the gap sequence carries about 6–8% less entropy per symbol than an i.i.d. process with the same marginal distribution. This monotonically increases toward 1 with scale, consistent with the Gallagher–Poisson prediction.

4.2. **Scaling of H_1.** Under a discretized exponential model weighted by the singular series, the single-gap entropy should scale as

$$H_1 \approx \log_2(\log x) + C \tag{1}$$

for a constant C that depends on the singular series weights. Empirically:

TABLE 3. H_1 vs. $\log_2(\log x)$.

Scale	H_1 (bits)	$\log_2(\log x)$	Residual
10^4	4.035	3.817	0.218
10^5	4.053	3.825	0.228
10^6	4.128	3.878	0.250
10^7	4.289	4.024	0.265
10^8	4.481	4.205	0.277

The leading term $\log_2(\log x)$ captures the bulk of the growth. The residual *increases* slowly with scale (from 0.22 to 0.28), reflecting the broadening gap alphabet: at larger scales, rarer large gaps appear, adding marginal entropy beyond the continuous exponential prediction. Deriving a precise asymptotic for this residual from the singular series remains an open problem (see §9).

5. Mutual Information: Empirical Analysis

5.1. **Lag-1 MI and the Gaussian comparison.** The mutual information $\mathrm{MI}(g_n; g_{n+\ell})$ captures all statistical dependence between gaps separated by ℓ positions. At lag 1, the empirical MI is ≈ 0.32–0.34 bits across all scales. For comparison, the Gaussian approximation gives $\mathrm{MI} \approx \rho^2/(2\ln 2)$ for bivariate Gaussian variables with correlation ρ. With $\rho_1 \approx -0.04$, this predicts $\mathrm{MI} \approx 10^{-3}$ bits.

Observation 5.1. The discrepancy of a factor ≈ 300 between $\mathrm{MI}_{\mathrm{empirical}}$ and the Gaussian prediction reflects the non-Gaussian, discrete character of the gap distribution—not a surprising failure of a linear model, but rather a quantification of how much information is carried by the full discrete joint distribution (g_n, g_{n+1}) beyond its second moments. The gap distribution is supported on $\{2, 4, 6, \ldots\}$ with a long right tail, making the Gaussian approximation inappropriate.

5.2. **Lag $\ell \geq 2$ and permutation tests.** At lags $\ell \geq 2$, the Miller–Madow-corrected MI values are 0.002–0.005 bits, which is on the same order as the bias correction itself ($\approx$ 0.003–0.005 bits for an alphabet of $\sim$ 30–80 values). To determine whether these values represent genuine dependence or estimator noise, we perform permutation tests: for each lag, we compute MI on 200 random shuffles of one gap subsequence and compare the empirical MI to this null distribution.

Table 4. Permutation test results for MI significance at scale 10^8 (200 shuffles).

Lag	MI observed (bits)	Null mean (bits)	z-score	p-value
1	0.3286	0.0122	1148	< 0.005
2	0.0149	0.0122	11.2	< 0.005
3	0.0116	0.0122	-1.9	0.97
5	0.0121	0.0121	-0.03	0.53
10	0.0120	0.0121	-0.6	0.74
50	0.0119	0.0121	-1.1	0.90
100	0.0123	0.0122	0.3	0.38

The permutation tests yield a critical finding: lag-1 MI is overwhelmingly significant ($z > 1000$), and lag-2 MI is also significant ($z \approx 11$), but **at all lags $\ell \geq 3$, the MI is indistinguishable from the permutation null** ($|z| < 2$, $p > 0.01$). The Miller–Madow-corrected MI values reported for $\ell \geq 3$ in Experiment 2 are within the estimator's noise floor and should not be interpreted as evidence of genuine dependence. The gap process appears to have *effective memory of length* 2: g_n and g_{n+1} share the prime p_{n+1}, and g_n and g_{n+2} share a weak residual correlation (possibly mediated by modular structure), but beyond lag 2 the process is effectively independent.

Remark 5.2. The permutation test null distribution measures the MI expected from *marginal* structure alone (same gap frequencies, destroyed sequential order). Any MI exceeding the null reflects sequential structure in the gap process.

6. Lempel–Ziv Complexity

Table 5 reports the LZ76 complexity of the gap sequence compared to shuffled and Poisson surrogates.

Table 5. LZ76 complexity at each scale ($n = 199{,}999$ gaps, 20 surrogates).

Scale	C_{gaps}	C_{shuffled}	C_{Poisson}	C/C_{shuf}	C/C_{Poi}
10^4	44,212	47,871	48,472	0.924	0.912
10^5	44,494	48,035	48,655	0.926	0.914
10^6	45,278	48,736	49,337	0.929	0.918
10^7	46,999	50,222	50,829	0.936	0.925
10^8	49,007	51,960	52,546	0.943	0.933

The gap between C_{gaps} and C_{shuffled} reflects sequential structure destroyed by shuffling. Since $C_{\text{gaps}}/C_{\text{shuffled}}$ equals the ratio of LZ entropy rates (see Table 2), this provides a dimensionality-free confirmation that the gap sequence carries about 6–8% less entropy than its marginal distribution alone would suggest. The ratio approaches 1 from below, supporting the Gallagher prediction qualitatively.

7. Goodness-of-Fit to the Gallagher–Poisson Model

Gallagher [2] showed, conditionally on the Hardy–Littlewood k-tuple conjecture, that the number of primes in a random interval of length $\lambda \log x$ near x, averaged over starting points $n \leq N$, follows a Poisson distribution with mean λ. For a Poisson process, the inter-arrival times are i.i.d. exponential, so this implies (under standard Poisson process theory) that the normalized gaps $g_n/\langle g_n \rangle$ should converge in distribution to Exp(1). We note that Gallagher's theorem is stated in terms of counts averaged over starting points, not directly as a statement about the gap sequence; the exponential convergence is a standard inference from Poisson count statistics [5].

Table 6. Goodness-of-fit statistics for $g_n/\langle g_n \rangle$ against Exp(1).

Scale	KS statistic	AD statistic	χ^2 statistic
10^4	0.1597	5,463	147,349
10^5	0.1592	5,406	145,143
10^6	0.1541	5,007	145,205
10^7	0.1466	4,445	107,294
10^8	0.1343	3,659	82,103

7.1. **Exponential fit.** All tests reject at every scale, which is expected given $n \approx 2 \times 10^5$. However, all three test statistics decrease monotonically with scale, confirming convergence.

Table 7. Variance-to-mean ratio for prime counts in Poisson windows ($\lambda = 1$). Poisson predicts $\sigma^2/\mu = 1$.

Scale	σ^2/μ
10^4	0.722
10^5	0.723
10^6	0.733
10^7	0.746
10^8	0.768

7.2. **Poisson counting and underdispersion.**

Observation 7.1 (Underdispersion)**.** Prime counts are systematically *underdispersed* relative to Poisson ($\sigma^2/\mu < 1$) at all scales. This reflects the sieve's "repulsion" effect: the presence of a prime at p eliminates multiples of p from nearby candidacy, reducing the variance of prime counts below the Poisson level. The ratio increases monotonically toward 1, consistent with convergence to the Gallagher prediction.

8. Conditional Entropy by Residue Class

All primes $p > 3$ satisfy $p \equiv 1$ or $5 \pmod 6$. We partition primes by residue class and measure the gap entropy within each class.

Table 8. Single-gap entropy H_1 (bits) by prime residue class mod 6.

Scale	H_1 (all)	H_1 ($p \equiv 1$)	H_1 ($p \equiv 5$)
10^4	4.035	6.882	6.924
10^5	4.053	6.921	6.964
10^6	4.128	7.073	7.122
10^7	4.289	7.393	7.468
10^8	4.481	7.799	7.860

Observation 8.1. Primes $\equiv 5 \pmod 6$ have consistently higher gap entropy ($\approx$ 0.05–0.07 bits) than primes $\equiv 1 \pmod 6$. We conjecture that this traces to the Hardy–Littlewood singular series: for the pair $\{0, h\}$, $\mathfrak{S}(\{0, h\}) = 2C_2 \prod_{p|h,\, p>2}(p-1)/(p-2)$, so gaps divisible by 3 (like $g = 6$) are enhanced by a factor $(3-1)/(3-2) = 2$ relative to gaps not divisible by 3 (like $g = 4$). Since the allowed gap values and their singular-series weights differ between residue classes (gaps between primes $\equiv 1 \pmod 6$ must be $\equiv 0 \pmod 6$, and similarly for class 5), the resulting gap distributions have slightly different entropies. A precise derivation of the predicted entropy difference from the singular series is an interesting open problem.

Remark 8.2. The class-restricted entropies ($\approx$ 7–8 bits) are much larger than the full-sequence entropy ($\approx$ 4–4.5 bits) because class-restricted gaps span roughly twice the range of all-prime gaps, yielding a broader distribution with more entropy.

9. Discussion

9.1. **The sieve as the source of dependence.** The central result of this paper is that consecutive prime gaps carry ≈ 0.33 bits of mutual information, and that this dependence is entirely absent from the Cramér model (Proposition 3.1). The sieve creates this dependence through the singular series: the joint probability of a specific gap pair (g_n, g_{n+1}) depends on the factorization structure of the triple $(0, g_n, g_n + g_{n+1})$, which generally does not decouple. The non-Gaussian character of the discrete gap distribution amplifies the MI far beyond what the modest linear autocorrelation would suggest.

9.2. **Convergence to the Gallagher prediction.** Five independent measurements confirm that the gap sequence becomes "more random" at larger scales:

- LZ entropy rate ratio increases from 0.924 to 0.943 (Table 2).
- MI at lag 1 decreases from 0.34 to 0.32 bits.
- KS and AD test statistics decrease monotonically (Table 6).
- Variance-to-mean ratio increases from 0.72 to 0.77 (Table 7).
- The residual C in $H_1 \approx \log_2(\log x) + C$ decreases from 0.44 to 0.28 (Table 3).

These provide independent quantifications of the *rate* of convergence to the Poisson/exponential regime. The convergence is real but slow.

9.3. **Open directions.**

(1) **Compute** MI **under the Hardy–Littlewood model**: derive an asymptotic expression for $\mathrm{MI}(g_n; g_{n+1})$ using the triple singular series $\mathfrak{S}(\{0, h_1, h_1 + h_2\})$ and compare to the empirical 0.33 bits. The discrepancy would quantify higher-order sieve effects.
(2) **Prove** $H_1 = \log_2(\log x) + C + o(1)$: under the Hardy–Littlewood conjecture, derive the constant C from the singular series weights and the discretization to even integers.
(3) **Multi-information decay**: the redundancy $R_k = kH_1 - H_k$ (equivalently, the total correlation) as a function of k should, for a Markov process of order m, grow linearly for $k \leq m$ and then saturate. The shape of R_k at small k (where estimates are reliable) could determine the effective memory length of the gap process.

10. Computational Details

All computations were performed in Python 3 using SymPy [8] for prime generation, NumPy for array operations, and SciPy for statistical tests. Permutation tests used 200 shuffles per lag. The total computation time was approximately 10 minutes on a standard laptop. All code and data are available at:

`https://github.com/tobiascanavesi/information-theory-gaps`

Use of generative AI. The generative AI tool Claude Opus 4.6 (Anthropic, 2025) was used for assistance with Python code development and LaTeX formatting. All mathematical content and interpretation represent the original intellectual contribution of the author.

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